

```
public static int task3CM(int[] input) {  
    int maxLength = 0;  
    int sum = 0;  
    HashMap<Integer, Integer> sumMap = new HashMap<>();  
    for (int i = 0; i < input.length; i++) {  
        sum += input[i];  
        if (sumMap.containsKey(sum)) {  
            maxLength = Math.max(maxLength, i - sumMap.get(sum));  
        } else {  
            sumMap.put(sum, i);  
        }  
    }  
    return maxLength;  
}  
  
public static void main(String[] args) {  
    int[] input = {15, -2, 2, -8, 1, 7, 10, 23};  
    int result = task3CM(input);  
    System.out.println(result);  
}
```